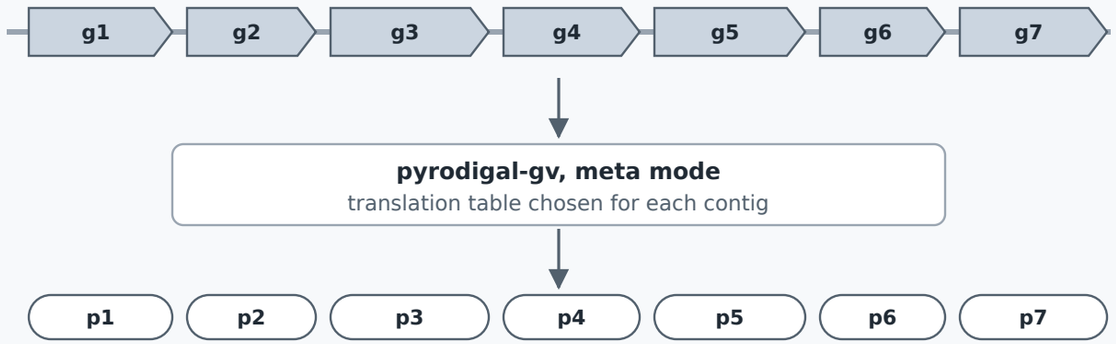
phu avger: finding auxiliary viral gene (AVG) candidates in viral contigs

A Gene prediction

Trusted viral contig (FASTA)



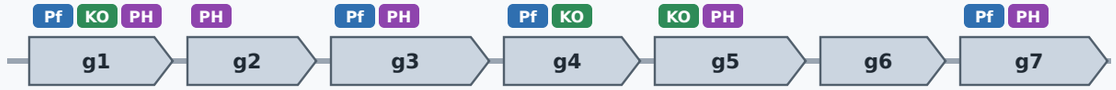
Translated proteins (at least 30 aa by default)

Predictions are cached and shared with phu screen and phu jack

B Two HMM annotation tracks (PyHMMER)

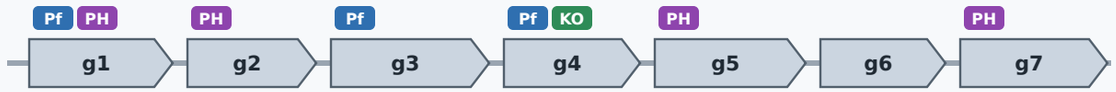
Pfam · KOfam · PHROG

Relaxed track E-value ≤ 1e-5; best hit per protein in each database



→ feeds V / VL scoring (panel C)

Strict track Pfam GA cutoffs · KOfam KO thresholds · PHROG bit score ≥ 60, coverage ≥ 0.4

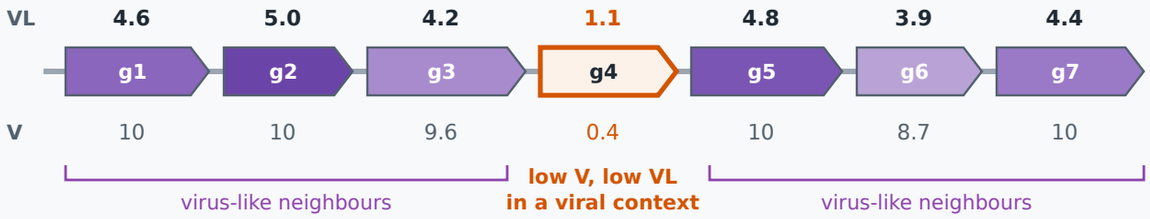


→ feeds functional calls and reference matching (panel D)

Pfam, KOfam and PHROG are scored and averaged independently; values are never mixed

C Viral-likeness scores (V-Score-Search tables)

Relaxed-track hits are looked up in the V / VL table, separately for each database



contig AVL (per database) = mean VL of the contig's scored genes

here: (4.6 + 5.0 + 4.2 + 1.1 + 4.8 + 3.9 + 4.4) / 7 = 4.0

V-score how exclusively a protein family occurs in viruses (10 = virus only)

VL-score V-score weighted by how often the family is found in viruses

AVL average VL across a contig, per database: the viral context of a gene

VL shading: lower … higher below the gene cutoff

D Candidate test and curation

Candidate test, per database

contig AVL	**> 3**	Pfam, KOfam and PHROG are each tested on their own
gene VL	**< 3**	
gene V	**< 10**	

fails in all → **not_candidate**

passes in at least one database → **AVG candidate**

1. **Match strict hits to the CheckAMG positive lists**
   AMG: Pfam, KOfam · APG and AReG: Pfam, KOfam, PHROG
2. **AMG weight ≥ 0.6 (--min-amg-weight)**
   applies to AMG evidence only
3. **Class filters (--filter-mode standard | strict | none)**
   an essential or structural hit in any database, PHROG included, removes the class

Final status

classified: putative_amg · putative_apg · putative_areg

class_conflict · filtered · below_amg_weight
unclassified_candidate · not_candidate

Outputs

gene-annotations.tsv	avg-candidates.tsv	avg-predictions.tsv	avg-evidence.tsv	.phu/run.json
every predicted gene	genes passing the test	one final named class	hits and reference matches	parameters and database versions

All calls are computational candidates for prioritisation, not confirmed auxiliary genes.

Scores: Zhou et al. 2026 · reference tables: CheckAMG · terminology: Martin et al. 2025 · gene values shown are illustrative